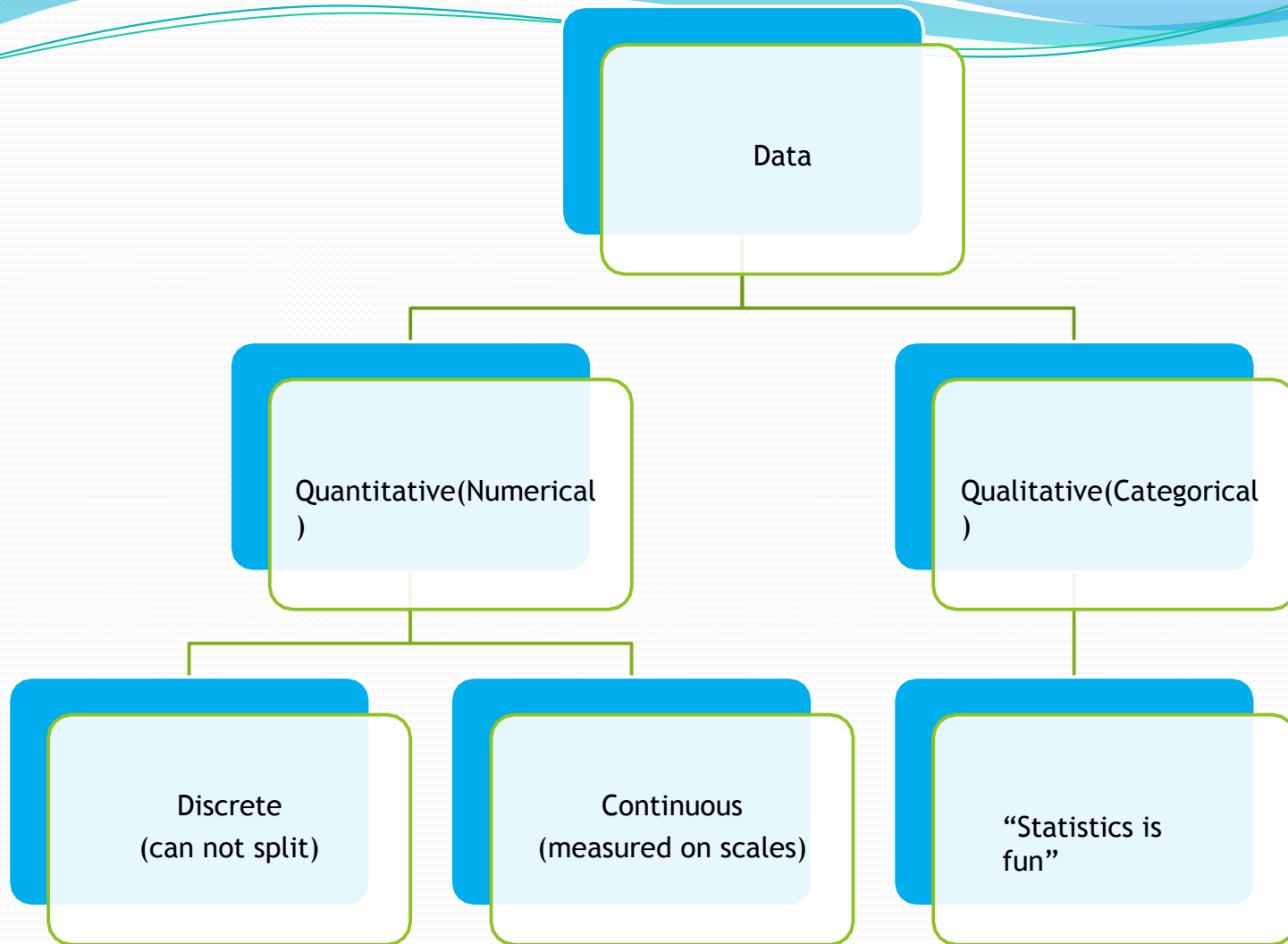


Discrete Structures

Instructor: Engr. Dua Agha

What is Discrete Structures?

- Discrete Structures is a branch of mathematics that studies **discrete (separate and distinct) objects** rather than continuous values.
- **Importance in Computer Science?**
 - ▶ Computer is a binary machine and all the algorithms in computer science are based on binary digits 0 and 1. We therefore can say computer is inherently is **DISCRETE**.



3

2.3434

What is qualitative data?

This bookcase...

- Is made of wood
- Was built in Italy
- Is deep brown
- Has golden knobs
- Smells like oak
- Has a smooth finish



What is quantitative data?

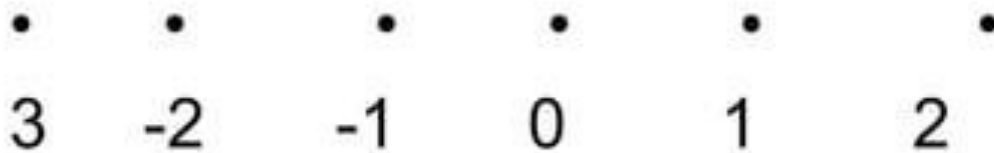
This bookcase...

- Is 3 feet tall
- Weighs 100 pounds
- Has 15 books on it
- Has 3 shelves
- Has 2 cabinets
- Sells for \$1500

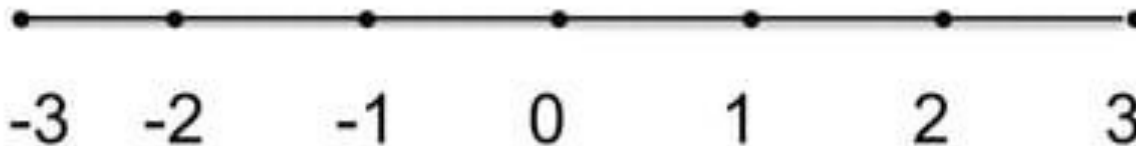


Example:

► Set of Integers:



► Set of Real Numbers:



► **Discrete Data : (at a given time)**

- ❖ Number of boys in a class 9
- ❖ Number of candies in a packet 10
- ❖ Number of bags lost by an airline 4



Continuous data : (at a give time)

- ❖ Height of a person 5ft 3 inches
- ❖ Time in a race 2 mins 40 sec
- ❖ Distance Travelled by a car 2.3 km

Sets

Sets

- A *set* is an unordered collection of objects.
 - the students in this class
 - the chairs in this room
- The objects in a set are called the *elements*, or *members* of the set.
- Sets are usually denoted by **capital letters: A, B, C**
- The notation **$a \in A$** denotes that **a is an element of the set A .**
- If **a is not a member of A** , write **$a \notin A$**

Describing a Set: Roster Method

- $S = \{a, b, c, d\}$

- **Order not important**

$$S = \{a, b, c, d\} = \{b, c, a, d\}$$

- **Each distinct object is either a member or not**; listing more than once does not change the set.

$$S = \{a, b, c, d\} = \{a, b, c, b, c, d\}$$

- **Elipses (...)** may be used to describe a set without listing all of the members **when the pattern is clear**.

$$S = \{a, b, c, d, \dots, z\}$$

Roster Method

- Set of **all vowels** in the English alphabet:

$$V = \{a, e, i, o, u\}$$

- Set of **all odd positive integers** less than 10:

$$O = \{1, 3, 5, 7, 9\}$$

- Set of **all positive integers less than 100**:

$$S = \{1, 2, 3, \dots, 99\}$$

- Set of **all integers less than 0**:

$$S = \{\dots, -3, -2, -1\}$$

Some Important Sets

$\mathbf{N}$ = *natural numbers* = $\{1, 2, 3, \dots\}$

$\mathbf{Z}$ = *integers* = $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

$\mathbf{Z}^+$ = *positive integers* = $\{1, 2, 3, \dots\}$

$\mathbf{R}$ = set of *real numbers*

● **Rational numbers** – numbers that can be written as fractions ($1/2, -3/4, 7, 0.25$)

● **Irrational numbers** – numbers that cannot be written as fractions, and their decimal goes on forever without repeating ($\sqrt{2}, \pi, e$)

$\mathbf{R}^+$ = set of *positive real numbers*

The set of $\mathbf{R}^+$ represents all **positive real numbers**, meaning any real number strictly greater than zero, excluding zero and negative numbers, with examples including decimals (0.5, 3.14), fractions ($1/2, 3/4$), integers (1, 5, 100), and irrational numbers like $\pi, \sqrt{2}$

$\mathbf{C}$ = set of *complex numbers*.

$$\mathbf{Z} = a + bi$$



► A set can be defined by simply listing its members inside curly braces. For example, the set $\{2, 4, 17, 23\}$ is the **same** as the set $\{17, 4, 23, 2\}$.

► To denote membership we use the $\in$ symbol, as in $4 \in \{2, 4, 17, 23\}$.

► On the other hand, non-membership is denoted as in $5 \notin \{2, 4, 17, 23\}$.

► Sets can be represented in three forms:

1. **Tabular Form.**
2. **Descriptive Form.**
3. **Set Builder Form.**

TABULAR FORM

- ▶ Listing all the elements of a set, separated by commas and enclosed within braces or curly brackets {}.

EXAMPLES

In the following examples we write the sets in tabular form.

- ▶ $A = \{1, 2, 3, 4, 5\}$
- ▶ $B = \{2, 4, 6, 8, \dots, 50\}$
- ▶ $C = \{1, 3, 5, 7, 9, \dots\}$

NOTE

- ▶ The symbol “...” is called an ellipsis. It is a short for “**and so forth.**”

DESCRIPTIVE FORM

Stating in words the elements of a set.

EXAMPLES

Now we will write the same examples which we wrote in tabular form, in the descriptive form.

- ▶ **A = set of FIRST FIVE NATURAL NUMBERS.**
- ▶ **B = set of POSITIVE EVEN INTEGERS LESS OR EQUAL TO FIFTY.**
- ▶ **C = set of POSITIVE ODD INTEGERS.**

SET BUILDER FORM

- ▶ Writing in **symbolic form** the common characteristics shared by all the elements of the set.
- ▶ The general form of this notation is $\{x \mid x \text{ has property } P\}$ and is read “**the set of all x such that x has property P.**”
- ▶ For instance, the set O of all odd positive integers less than 10 can be written as

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\},$$

or,

- ▶ Specifying the universe as the set of positive integers, as

$$O = \{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 10\}.$$

EXAMPLES

Now we will write the same examples which we wrote in tabular as well as descriptive form ,in set builder form .

- ▶ $A = \{x \in \mathbb{N} \mid x \leq 5\}$ (**is the Set Builder Form**)
- ▶ $B = \{x \in \mathbb{E} \mid 0 < x \leq 50\}$ (**is the Set Builder Form**)
- ▶ $C = \{x \in \mathbb{O} \mid 0 < x \}$ (**is the Set Builder Form**)

Example 1.

- 1. Write the set of natural numbers that is less than 6 in roster notation and in set builder notation?

Solution:

- Roster notation

$$A = \{1, 2, 3, 4, 5\}$$

- Set builder notation

- $A = \{x \mid x \in \mathbb{N}, 1 \leq x \leq 5\}$

Example 2.

- 2. Write the set of whole numbers that is less than 8 in roster and in set builder notation

Solution:

Roster notation

$$B = \{0, 1, 2, 3, 4, 5, 6, 7\}$$

Set builder notation

$$B = \{x \mid x \in W, 0 \leq x < 8\}$$



Interval Notation

$$[a,b] = \{x \mid a \leq x \leq b\}$$

$$[a,b) = \{x \mid a \leq x < b\}$$

$$(a,b] = \{x \mid a < x \leq b\}$$

$$(a,b) = \{x \mid a < x < b\}$$

closed interval $[a,b]$

open interval (a,b)

Universal Set

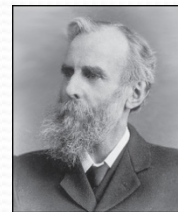
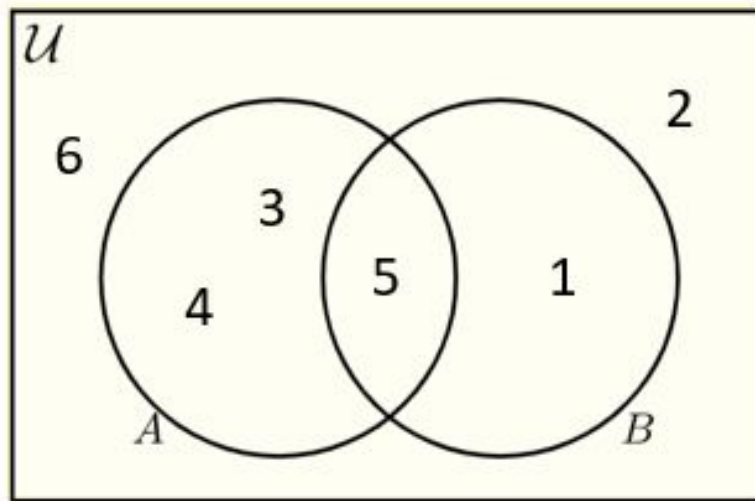
- The universal set is the set of all elements under consideration, denoted by capital U .

Example:

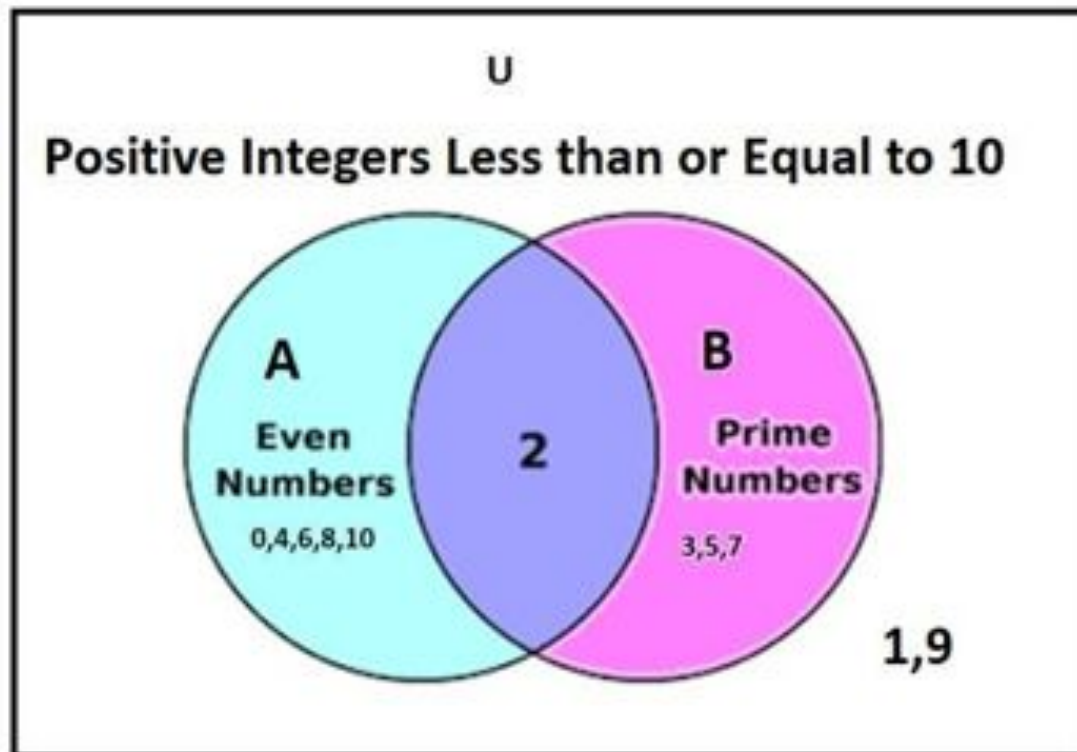
$$\mathcal{U} = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{3, 4, 5\}$$

$$B = \{1, 5\}$$



John Venn (1834-1923)
Cambridge, UK



U = Set of all positive Integers less than or equal to 10.

A = Set of even numbers.

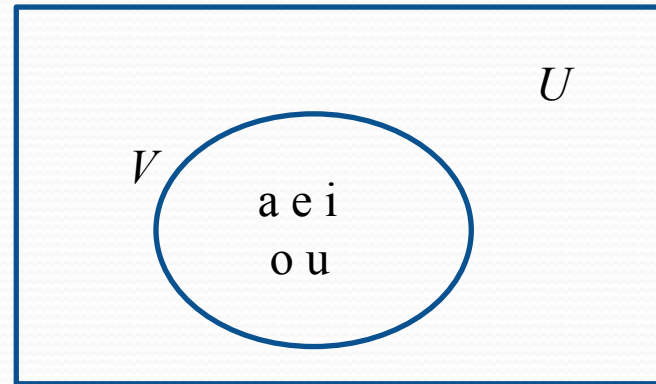
B = Set of prime numbers

U = {0,1,2,3,4,5,6,7,8,9,10}.

A = {0,2,4,6,8,10}

B = {2,3,5,7}

Universal Set



Empty Set

A set with **no elements**.

- Symbol: $\emptyset$ or $\{\}$

NOTE: $\emptyset \neq \{\emptyset\}$

- $\emptyset$ has 0 elements
- $\{\emptyset\}$ has 1 element

Set Equality

Definition: Two sets are *equal* if and only if they have the same elements.

- Therefore if A and B are sets, then A and B are equal if and only if $\forall x (x \in A \leftrightarrow x \in B)$
- We write $A = B$ if A and B are equal sets.

$$\{1, 3, 5\} = \{3, 5, 1\}$$

$$\{1, 5, 5, 5, 3, 3, 1\} = \{1, 3, 5\}$$

Subsets

Definition: The set A is a *subset* of B , **if and only if every element of A is also an element of B .**

- The **notation $A \subseteq B$** is used to indicate that A is a subset of the set B .
- $A \subseteq B$ holds if and only if $\forall x(x \in A \rightarrow x \in B)$ **is true.**

Example

- $A = \{1, 3, 5\}$, $B = \{1, 2, 3, 4, 5\}$, $C = \{1, 2, 3, 4\}$
- $A \subseteq B$ (Because every element of A is in B)
- $C \subseteq B$ (Because every element of C is also an element of B)

Showing a Set is or is not a Subset of Another Set

- **Showing that A is a Subset of B:** To show that $A \subseteq B$, show that if x belongs to A , then x also belongs to B .
- **Showing that A is not a Subset of B:** To show that A is not a subset of B , $A \not\subseteq B$, find an **element $x \in A$ with $x \notin B$** . (Such an x is a counterexample to the claim that $x \in A$ implies $x \in B$.)

Example:

- Every **computer science (CS) major** is also a **student of the school**.
- **Why this is TRUE:**
 - **A = set of all CS majors**
 - **B = set of all students at the school**
 - Anyone who is a CS major **must be enrolled in the school**
 - There cannot be a CS major who is *not* a student of the school

Another look at Equality of Sets

- Recall that two sets A and B are *equal*, denoted by $A = B$, iff

$$\forall x (x \in A \leftrightarrow x \in B)$$

- Using **logical equivalences** we have that $A = B$ iff

- $\forall x [(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)]$
This is equivalent to

$$A \subseteq B \quad \text{and} \quad B \subseteq A$$

Example:

$$\{1,3,5\} = \{5,1,3\}$$

Proper Subsets

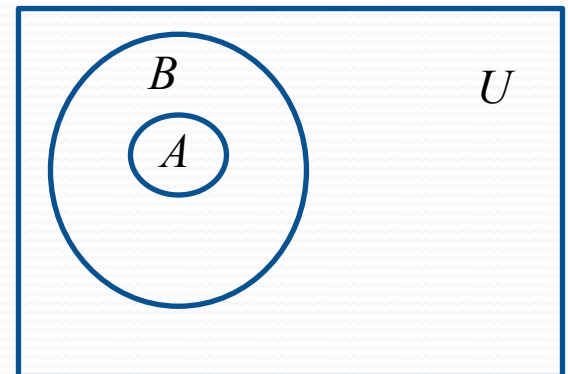
Definition: If $A \subseteq B$, but $A \neq B$, then we say A is a *proper subset* of B , denoted by $A \subset B$.

Let A and B be sets. **A is a proper subset of B** , if, and only if, **every element of A is in B but there is at least one element of B that is not in A** , and is denoted as **$A \subset B$** .

EXAMPLE Let $A = \{1, 3, 5\}$, $B = \{1, 2, 3, 5\}$

● then $A \subset B$ (Because there is an element 2 of B which is not in A).

Venn Diagram



Set Cardinality

The cardinality of a set is the number of distinct elements in it.

Notation: $|A|$

Examples:

1. $|\emptyset| = 0$
2. Let S be the letters of the English alphabet. Then $|S| = 26$
3. $|\{1,2,3\}| = 3$
4. $|\{\emptyset\}| = 1$
5. The set of integers is infinite.

Finite vs Infinite Sets **Finite:** $\{1,2,3\}$, **Infinite:** $\mathbb{Z}$ (integers)

Power Sets

Definition: The set of all subsets of a set A , denoted $P(A)$, is called the *power set* of A .

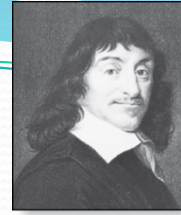
Example: If $A = \{a, b\}$ then

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

- If a set has n elements, then the **cardinality of the power set is 2^n** . (In Chapters 5 and 6, we will discuss different ways to show this.)

Tuples

- The *ordered n-tuple* $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element and a_2 as its second element and so on until a_n as its last element.
- Two n-tuples are equal if and only if their corresponding elements are equal.
- 2-tuples are called *ordered pairs*.
- The ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$.
- **Examples:**
 - $(2, 5) = (2, 5)$ ✓
 - $(2, 5) \neq (5, 2)$ ✗



René Descartes
(1596-1650)

Cartesian Product

Definition: The *Cartesian Product* of **two sets A and B** , denoted by **$A \times B$** is the set of ordered pairs (a,b) where $A \times B = \{(a,b) \mid a \in A \text{ and } b \in B\}$.

Example:

$$A = \{a,b\} \quad B = \{1,2,3\}$$

$$A \times B = \{(a,1),(a,2),(a,3), (b,1),(b,2),(b,3)\}$$

Cartesian Product

Definition: The cartesian products of the sets $A_1, A_2, \dots, A_n$, denoted by $A_1 \times A_2 \times \dots \times A_n$, is the set of ordered n -tuples $(a_1, a_2, \dots, a_n)$ where a_i belongs to A_i for $i = 1, \dots, n$.

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}$$

Example: What is $A \times B \times C$ where $A = \{0,1\}$, $B = \{1,2\}$ and $C = \{0,1,2\}$

Solution: $A \times B \times C = \{(0,1,0), (0,1,1), (0,1,2), (0,2,0), (0,2,1), (0,2,2), (1,1,0), (1,1,1), (1,1,2), (1,2,0), (1,2,1), (1,2,2)\}$